

Introduction to Social Data Analytics: Week 2 Part 2

Today

- ▶ Introduce generate command
- ▶ Logical statements in Stata
- ▶ Introduce histograms (theory) and how to construct in Stata
- ▶ Construct a histogram that informs us about variation in mobility rates across colleges in the US

Setup: loading the data

- ▶ Reminder, we need to change the working directory to the location of our data and then load our data

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/02_week/data"  
use college_mobility.dta, replace
```

(Preferred Estimates of Access and Mobility Rates by College)

Describe

- The describe command allows us to quickly review the variables in our dataset

describe

Contains data from **college_mobility.dta**

Observations: **2,199**

Variables: **7**

**Preferred Estimates of Access and
Mobility Rates by College**

21 Jul 2021 13:14

Variable name	Storage type	Display format	Value label	Variable label
name	str141	%40s		Name of Institution / Super-OPEID Cluster
state	str2	%9s		State of constituent institution with highest enrollment
par_median	double	%9.0g		Median parental income
k_median	float	%9.0g		Median kid income
par_q1	float	%9.0g		Fraction of parents in quintile 1 (bottom quintile)
kq5_cond_parq1	float	%9.0g		Probability of kid in quintile 5 conditional on parent in quintile 1
count	double	%12.0g		Mean number of kids per cohort

Browse

- ▶ `describe` gives us a good sense of what variables are in the data
- ▶ Many times it is also useful look directly at the data
- ▶ The `browse` command opens up a spreadsheet similar to an Excel spreadsheet
- ▶ This allows you to look through the data to further understand how the values of each variable are stored

Data Editor (Browse) — college_mobility.dta

par_q1[7]

.10234574

	name	state	par_median	k_median	par_q1	kq5_cond_p~1	count	
1	Vaughn College Of Aeronautics And Techn...	NY	30900	53000	.3647788	.4484354	207.6666667	
2	CUNY Bernard M. Baruch College	NY	42800	57600	.2763224	.4682423	1083	
3	City College Of New York - CUNY	NY	35500	48500	.3254648	.3602156	582.3333333	
4	CUNY Lehman College	NY	32500	40700	.3670749	.2788297	468.3333333	
5	California State University, Los Angeles	CA	36600	43000	.3311693	.2994981	1179.666667	
6	CUNY John Jay College Of Criminal Justi...	NY	41800	45200	.2715895	.3568414	1228	
7	MCPHS University	MA	83300	112700	.1023457	.9129356	173.5	
8	Pace University	NY	68600	60700	.1517329	.5557559	1353.333333	
9	State University Of New York At Stony B...	NY	73600	60100	.1641662	.512453	2070.666667	
10	New York City College Of Technology Of ...	NY	33500	37000	.3525651	.236384	1488.333333	
11	ITI Technical College	LA	56000	55000	.2025736	.4016365	92	
12	Texas A&M International University	TX	35400	42800	.3189379	.2535727	258	
13	CUNY Brooklyn College	NY	52200	44300	.23247	.346963	1009.666667	
14	Technical Career Institutes	NY	29700	31300	.403024	.1984375	562	
15	University Of Texas - Pan American	TX	31700	39300	.387026	.1976301	1723.666667	
16	CUNY Hunter College	NY	49800	44400	.2116361	.3563665	1489.333333	
17	Crimson Technical College	CA	44700	37600	.2503355	.3004882	156	
18	Wood Tobe - Coburn School	NY	29500	24500	.4131257	.1770672	129	
19	United Talmuddical Seminary	NY	21200	15300	.6097748	.1177	218	
20	CUNY Queens College	NY	63300	48200	.2014274	.354384	1052.666667	
21	Glendale Community College	CA	40100	30500	.3235921	.2188597	1437	
22	South Texas College	TX	23900	27500	.523607	.1319543	1881.333333	
23	California State Polytechnic University...	CA	80200	55100	.1492972	.4577048	2195.333333	
24	University Of Texas At El Paso	TX	42400	38400	.2800138	.2435862	1665.666667	
25	CUNY York College	NY	36500	36400	.3066044	.2224203	368	
26	Saint John's University of Queens, NY	NY	69200	58900	.1434363	.4737947	2363.333333	
27	University Of California, Irvine	CA	92100	60400	.1224691	.5526469	3244.333333	
28	Trocaire College	NY	65400	38200	.2365247	.2847948	82.5	
29	Western Technical College of El Paso, TX	TX	30500	23900	.4369181	.1541342	224	
30	Laredo Community College	TX	27400	28700	.4307666	.155818	1331.333333	
31	University Of Texas At Brownsville	TX	26400	29800	.4734971	.1403364	1351.333333	
32	Saint Francis College	NY	70500	53000	.1345287	.4920357	365.3333333	
33	CUNY Hunter College	NY	49800	44400	.2116361	.3563665	1489.333333	

Variables

Name	Label
☒ name	Name of Institution...
☒ state	State of constituen...
☒ par_median	Median parental inc...
☒ k_median	Median kid income
☒ par_q1	Fraction of parents...
☒ kq5_cond_parq1	Probability of kid in...
☒ count	Mean number of ki...

Properties

◀

▶

Variables

Name	par_q1
Label	Fraction of parents in quintile 1
Type	float
Format	%0.0g
Value label	
Notes	

Data

Frame	default
Filename	college_mobility.dta
Label	Preferred Estimates of Access and

Vars: 7 Order: Dataset

Obs: 2,199

Filter: Off

vars: 7 Order: Dataset

Obs: 2,199

Descriptive Statistics

- ▶ Whenever we load a new dataset, it is important to understand our variables of interest
- ▶ One helpful way to understand our variables is through descriptive statistics
- ▶ A descriptive statistic is any statistic that quantitatively describes the variable (common ones are the mean, median, and standard deviation)
- ▶ In Stata, the `summarize` command is useful for retrieving summary statistics

Summarizing the size of institutions

- ▶ The variable `count` is the average number of students enrolled in the institution
- ▶ To retrieve summary statistics for this variable we type

```
summarize count
```

Variable	Obs	Mean	Std. dev.	Min	Max
count	2,199	946.5153	1508.825	50	26989.67

Mobility Rates

- ▶ We want to construct the **mobility rate** for every college

$\text{mobility rate} = (\text{fraction of students from Q1}) \times (\text{fraction of Q1 students that reach Q5})$

- ▶ In our dataset, we have variables that capture the first term (`par_q1`) and the second term (`kq5_cond_parq1`)
- ▶ From these we can construct a college's mobility rate. To do this we will use the `generate` command

Basic syntax of the generate command

- ▶ The most basic syntax of the generate command is

```
gen newvar = exp
```

- ▶ `newvar` is the name of the new variable that you would like to create
 - ▶ You can give the variable any name you like, but generally it is good practice to keep names short and as descriptive as possible
- ▶ `exp` is an expression. How you fill in `exp` depends on what variable you are creating

Examples of generate syntax

- ▶ If we have two variables x_1 and x_2 , we can create new variables that are functions of these two variables
 - ▶ Addition: `gen sum = x1 + x2`
 - ▶ Subtraction: `gen diff = x1 - x2`
 - ▶ Multiplication: `gen mult = x1 * x2`
 - ▶ Division: `gen div = x1/x2`

Generating mobility rates

- ▶ To generate a new variable named `mobility_rate` we can type

```
gen mobility_rate = par_q1*kq5_cond_parq1
```

- ▶ If we additionally want to give this new variable a label we type

```
label var mobility_rate "Mobility rate of institution"
```

Logical statements

- ▶ A logical statement is any statement that is either true or false
- ▶ For example, the statement “College X is in California’ ’ is either true or it is false
- ▶ For “UCSD” this is true
- ▶ For “ASU” this is false

Creating binary variables

- ▶ It is often useful to create binary variables (binary meaning two) in our data based on logical statements
- ▶ For example, we can create a binary variable that is equal to 1 if a logical statement is true and zero if false

```
gen newvar = (Logical Statement)
```

- ▶ `newvar` is the name of the binary variable. It will be equal to 1 if the logical statement is true and zero otherwise

Creating binary variables (an example)

- ▶ Now let's create a binary variable that is equal to 1 if the college is in California and zero otherwise
- ▶ To do this in our college mobility dataset we type

```
gen CA = (state=="CA")
```

- ▶ Note the double-equals sign (==). We use double-equals to test the equality of two things
- ▶ Note the quotation marks around CA. state is a string variable (meaning it is composed of text) and we always reference string variables with quotation marks

Did we retrieve the expected results?

- ▶ To check whether the results conform to our expectations we can type `br state CA`
- ▶ Below is a small selection of the data that shows us CA is equal to 1 for colleges in “CA” and zero otherwise

15	TX	0
16	NY	0
17	CA	1
18	NY	0

Common mistakes (1/2)

- ▶ Let's see what happens if we forget to use double-equals

```
gen CA2 = (state="CA")
```

```
. gen CA2 = (state="CA")  
state="CA" invalid name  
r(198);
```

- ▶ The error report `state="CA"` is displayed because the command is looking for a variable with this name, which is not in a valid format
- ▶ To test equality, we need both equals signs

Common mistakes (2/2)

- ▶ Let's see what happens if we forget to use quotation marks when reference strings

```
gen CA3 = (state==CA)
```

```
. gen CA3 = (state==CA)
```

```
type mismatch
```

```
r(109);
```

- ▶ The error report `type mismatch` occurs because `state` is a string variable, yet what comes after the equals sign will NOT be interpreted as a string
- ▶ To tell Stata the characters `CA` are strings, put `CA` in quotation marks!

If statements

- ▶ If statements are used to run a line of code that conditions on a subset of observations
- ▶ For example, imagine we have a dataset of income
- ▶ We want to know the average income, but only for individuals older than 18
- ▶ In Stata, we could type

```
sum income if age>18
```

Comparisons

- ▶ Now compare UCSD to other schools
- ▶ To have Stata report the mobility rate for UCSD we can type

```
sum mobility_rate if name == "University Of California, San Diego"
```

Variable	Obs	Mean	Std. dev.	Min	Max
mobility_r~e	1	.0483275	.	.0483275	.0483275

- ▶ The mobility rate for UCSD (0.048) is much higher than the average in the state (0.028)
- ▶ We could keep doing comparisons like this, but doing so would be inconvenient. Next let's learn a visualization technique that will allow us to see all the variation in college mobility rates

Histograms

- ▶ A **histogram** is a representation of the distribution of a **numeric** variable
- ▶ In our empirical application, we want to understand the distribution of mobility rates across colleges
- ▶ To keep things simple, let's go through a hypothetical example

The distribution of test scores

- ▶ A distribution you are all likely familiar with is a distribution of test scores
- ▶ Let's use a histogram to represent the distribution of test scores for the following 10 hypothetical students

$$\{65, 71, 75, 76, 80, 84, 85, 86, 90, 98\} \quad (1)$$

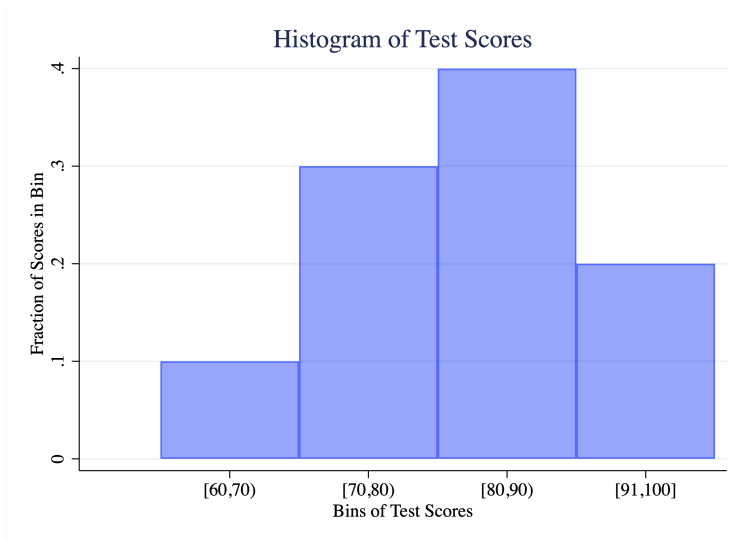
First step: “bin” the data

- ▶ The first thing we must decide is how to “bin” our data (sometimes referred to as “bucket”)
- ▶ Binning the data means to divide the range into a series of intervals
- ▶ For example, for test scores, I could use the following intervals
 - ▶ Interval 1: $[60,70)$
 - ▶ Interval 2: $[70,80)$
 - ▶ Interval 3: $[80,90)$
 - ▶ Interval 4: $[90,100]$
- ▶ Reminder: brackets mean inclusive, while parentheses mean excluding. So a score of 70 would fall in interval 2, not 1

Counting the fraction within each bin

- ▶ Next we compute the fraction of observations within each bin
- ▶ Only 1 test score (65) falls within interval 1, so 10 percent of observations are in interval 1
- ▶ Three test scores fall within interval 2, so 30 percent of observations are in interval 2
- ▶ We can compute the fraction in each interval and then plot the result in a histogram

The Histogram

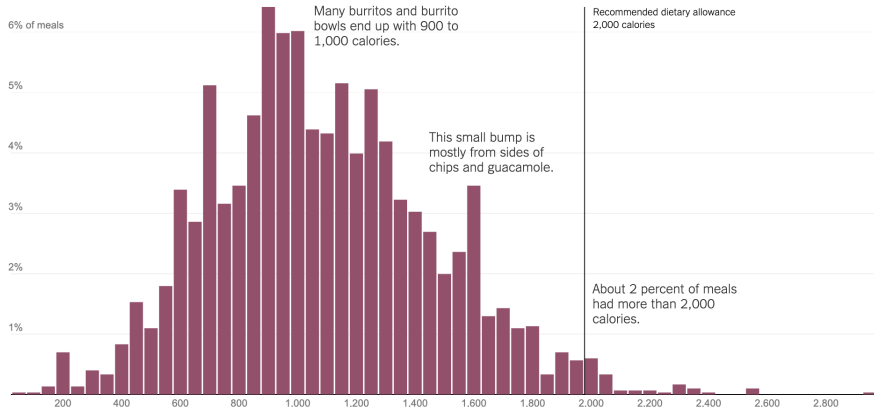


Another Histogram Example

At Chipotle, How Many Calories Do People Really Eat?

By KEVIN QUEALY, AMANDA COX and JOSH KATZ FEB. 17, 2015

Most meals have more than 1,000 calories and almost a full day's worth of sodium. [RELATED ARTICLE](#)



Histograms

- ▶ In our empirical application, we want to understand the distribution of mobility rates across colleges
- ▶ To begin, we need to learn about the `histogram` command
- ▶ The simplest possible syntax is to simply type

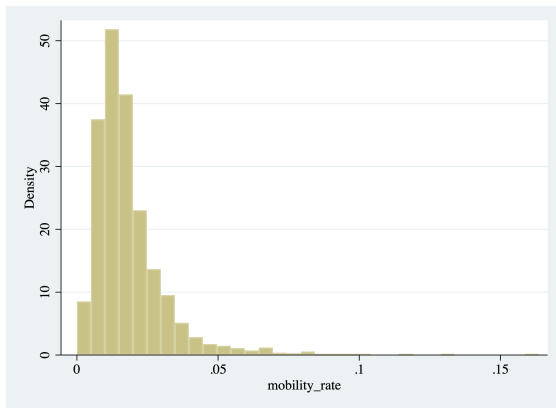
```
histogram varname
```

- ▶ where `varname` is the name of whatever variable you would like to create a histogram for
- ▶ In our example, we want to plot a histogram of `mobility_rate`

Histogram of mobility rates

- To plot a histogram of mobility rates we type

```
histogram mobility_rate
```



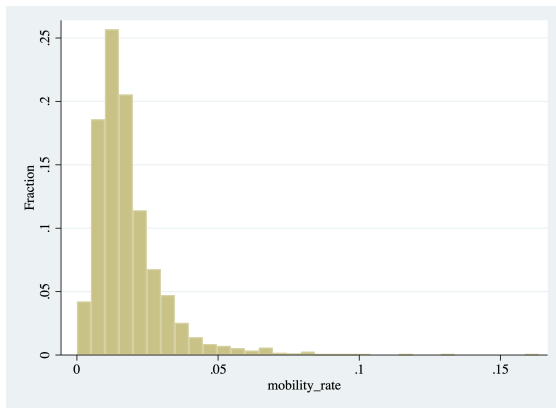
What is “Density”?

- ▶ You will notice that the y-axis says “density”, but in all our previous examples the y-axis was labeled “fraction of observations”
- ▶ We will mostly be plotting “fraction of observations” in this class
 - ▶ It is not necessary to understand this for the class, but a density integrates to 1, which is why the y-axis is sometimes rescaled in this way
- ▶ To plot fraction of observations instead of density, we can use an option!

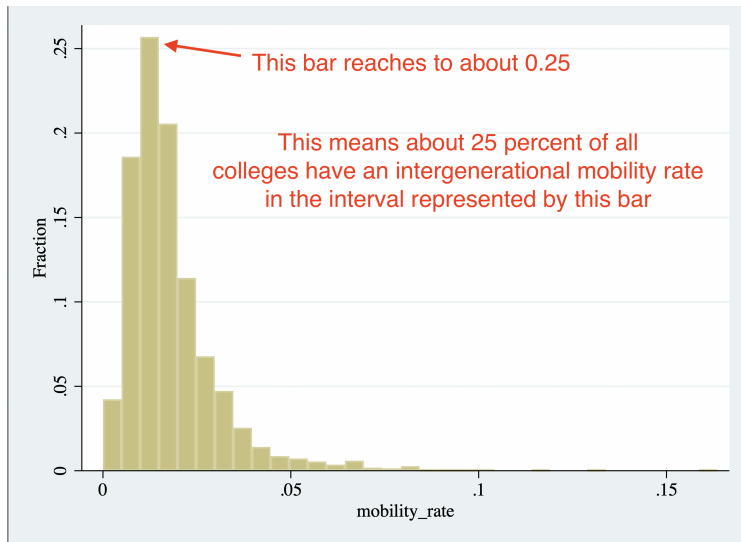
Histogram with Fraction of Observations

- To plot a histogram with fraction of observations as the y-axis type

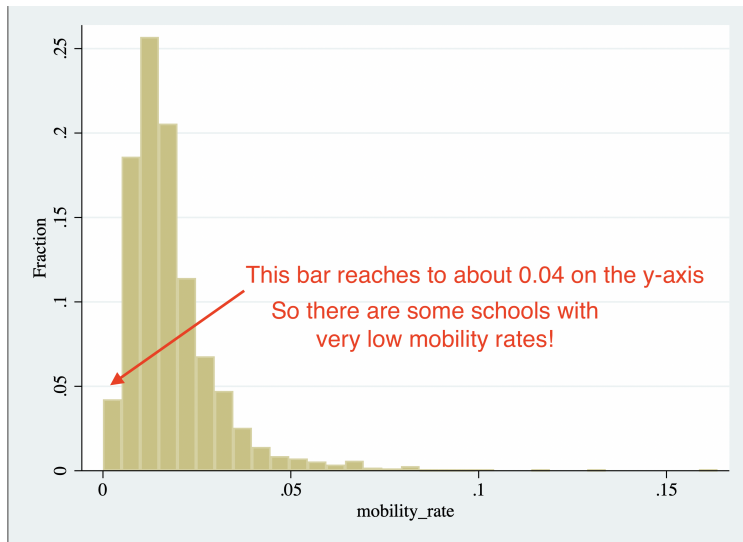
```
histogram mobility_rate, frac
```



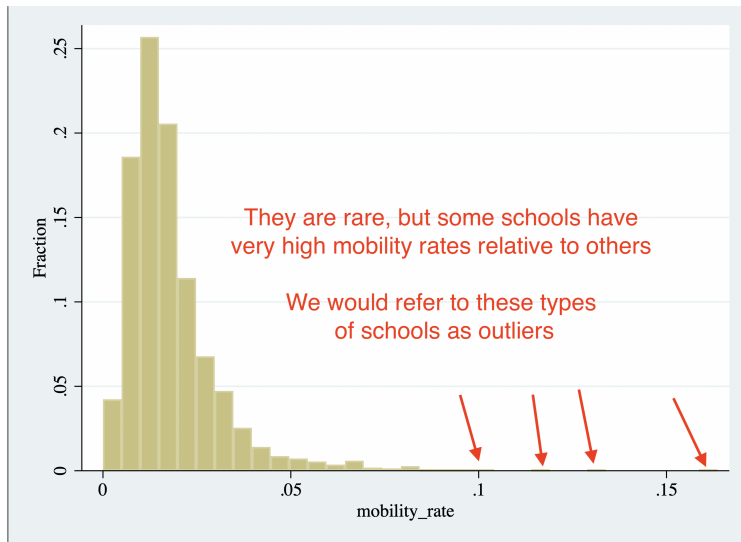
Interpreting the Histogram (1/3)



Interpreting the Histogram (2/3)

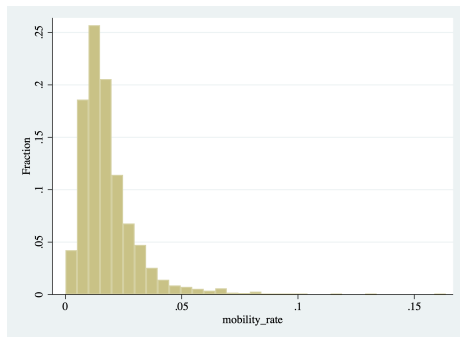


Interpreting the Histogram (3/3)



Our Histogram

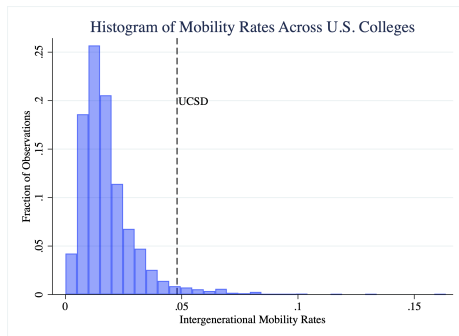
- ▶ This is what Stata outputs as the default format for a histogram



- ▶ This leaves a lot to be desired! Let's discuss ways to improve (1) informativeness of the graph and (2) aesthetics of the graph

Advanced graph

```
histogram mobility_rate, frac color(blue%40) ///  
  title("Histogram of Mobility Rates Across U.S. Colleges") ///  
  xtitle("Intergenerational Mobility Rates") ///  
  ytitle("Fraction of Observations") ///  
  xline(0.048, lcolor(black) lp(dash)) ///  
  text(0.2 0.055 "UCSD") ///  
  graphregion(fcolor(white))
```



What we have learned

- ▶ In this module, we studied a college's mobility rate
- ▶ We found wide ranges of mobility rates across U.S. colleges
 - ▶ Some colleges have close to zero, which can be due to either (1) very low access to low-income students or (2) low success rates of students
 - ▶ Some colleges (like UCSD) have relatively high mobility rates
- ▶ Researchers at Opportunity Insights have used this data to explore this mobility rates **extensively**
 - ▶ See Conclusions video on Canvas for more details about this

Key Takeaways from Opportunity Insights Research

- ▶ Access varies widely by parental income, but success does not
- ▶ Despite numerous policies aimed at increasing low-income enrollment at elite universities (e.g. Harvard), we do not see increases in low-income enrollment over time at these universities
- ▶ Many schools with very high access rates have been trending towards lower access in recent years
- ▶ Opportunity Insights suggests changing admissions criteria or increasing transfers from community colleges could be effective policies in promoting intergenerational mobility (among other potential policy solutions)